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Nonlinear fractional mathematical model of tuberculosis (TB) disease with incomplete treatment under Atangana-Baleanu derivative

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Abstract The current work investigates the mathematical model of Tuberculosis disease with incomplete treatment under the Atangana-Baleanu-Caputo (ABC) derivative with fractional order. Upon exploiting fixed point approach and nonlinear analysis, we derive some theoretical results about solution existence and its stability. The famous fractional Adam Bashforth technique is applied to compute numerical solution to the considered model. The aforesaid tool is based on fundamental theorem of fractional calculus and Lagrange interpolation polynomials. Additionally, various numerical plots are given corresponding to different fractional order in $(0, 1]$.

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1. Introduction

Different researchers have experimentally proven that TB (Tuberculosis) is one of the contagious epidemics that is usually caused by the bacteria of the lungs of human beings. According to WHO, in 2017 approximately 10 million persons were infected with TB disease and 1.50 million people have died from TB worldwide [1]. Researchers are anxious that a worldwide outbreak in the number of diabetes cases will place lots of individuals in danger [2].

To control the contagious disease, several mathematical models have been examined for the transmission dynamics and construction of useful methods. In disease epidemiology, the TB disease infection model was first presented by Waaler et al. [3]. Yang et al. [4] introduced a model to examine the part of various scenario of transmission of TB disease with incomplete treatment. Several researchers presented mathematical models for TB to examine its global stability results and the influence of heterogeneity on TB dispersion [5,6]. Zhang et al. [7] investigated a dynamical problem of TB disease including hospitalized and non hospitalized infectious classes. To investigate the consequences of deterioration and reinfection, a mathematical model of TB was considered by Robert [8]. Moreover, Egonmwan et al. [9] presented a model for TB disease to determine the influence of treatment and analysis for infectious persons.

The use of mathematical modeling helps in recognizing the transmission dynamics of each endemic, resulting in an understanding of how infection affects the new population. Ihsan et al. [10] formulate the stability of TB with incomplete treatment. The proposed model is given below;

$$\begin{cases} \frac{dS}{dt} = \Pi^* - \zeta^* S(\mathbb{I} + \zeta^* \mathbb{T}) - \varphi^* S, \\ \frac{dE}{dt} = \zeta^* S(\mathbb{I} + \zeta^* \mathbb{T}) - (\varphi^* + \lambda^*) E + (1 - \varrho^*) \delta^* \mathbb{T}, \\ \frac{dI}{dt} = \lambda^* E - (\varphi^* + \eta_1^* + \gamma^*) \mathbb{I}, \\ \frac{d\mathbb{I}}{dt} = \gamma^* \mathbb{I} - (\varphi^* + \eta_2^* + \delta^*) \mathbb{T}, \\ \frac{dR}{dt} = \varrho^* \delta^* \mathbb{T} - \varphi^* R. \end{cases} \quad (1)$$

The above TB mathematical model is divided into five groups namely, susceptible population $S(t)$, exposed population $E(t)$, infectious class without treatment $\mathbb{I}(t)$, infectious class with treatment $\mathbb{T}(t)$ and $R(t)$ is the recovered class. The involve various parameters in (1) are described as:

In previous twenty years, fractional-order (FO) models have gained more interest from the researcher. They deliver new precise and deeper facts around the complex action of several infections than classical integer-order models. Integer-order system are not higher than FO because of the genetic properties and description memory [11–18]. Mathematical models are frequently differential and integral equations of integer order (IDEs). Non-integer order fractional differential equations (FDEs) are used to improve the actual phenomena for a better degree of precision and accuracy. Furthermore, their utilization about fractional calculus are used and found in [19–25]. Researchers investigated and expended the classical calculus method to fractional-order derivative equations (FODEs) that are used in [26–29]. Several researchers used a different technique for examining the qualitative investigation of fractional order in modeling according to [30–38]. Generally, to find the general solution for the non-linear FDEs is not an easy task, therefore several researchers presented

various methods for the discovery of the approximate solution for nonlinear systems. Some dynamical problems have been handle via using various numerical techniques under fractional derivatives (for detail see [39–44,46]). Currently, several techniques are used to control DEs of fractional with classical order and their system such as the “residual power series technique, Fourier transform technique, spectral technique”, and technique for assemblage as well as many other procedures of computational approaches, etc. details are in [45,47] (see Table 1).

Recently, many researchers suggested several kinds of fractional order and non-local derivative that conclude the decrease of an integer-order derivative. For instance, Riemann–Liouville introduces the theory of fractional derivative, with further, modifications have done by Caputo. Using the definition of fractional derivatives singularity has revealed yields these types of operatives impractical for the dynamical mathematical models. For controlling these problems another mathematician Caputo-Fabrizio (CF) [48] recommended a novel FO derivative using the exponential function as their kernel. To understand the terminology for CF we add some application for details see, [49–51]. A novel concept was presented in the literature where the kernel was changed from non-local singular to non-local and non-singular. The advantage of this new kernel is the depiction of disappearing memory and well-defined memory of the system under examination. In 2016, Atangana and Baleanu [52] constructed a novel derivative, which generalized the Mittag–Leffler function to non-local and non-singular. This newly derivative has been applied in the mathematical modeling of different real-world problems in various fields [53–59,66]. Therefore, most of the researchers are interested in investigating the area of the biological model of infectious disease. Some investigation of the model are anxious about the optimization and stability existence, see [60–63].

In our work, our main support relates to the attention of the greatest general classification which currently looks in into medical journals [64]. This paper contains existence and uniqueness, practical analysis and numerical simulations on reviewing the dynamic behavior of the TB infection and in how to decrease the infection in human beings. Studying the model (1) in the form of FODEs under Attangana-Baleanu-Caputo $\mathbb{ABC}$ derivative of fractional order ℓ along with initial conditions is considered below as:

Table 1 Parameter with description occurring in the model (1).

Notation	Details of the parameter
Π^*	the rate of conscription
ϱ^*	capable treatment
φ^*	the rate of natural death
η_1^*	infected person without treatment $\mathbb{I}$
ζ^*	decrease in infectious
η_2^*	infected person with treatment $\mathbb{T}$
ζ^*	the rate of effective contact
γ^*	the rate for infected population $\mathbb{I}$
λ^*	the rate of transfer from E to $\mathbb{I}$
δ^*	the rate of leaving population $\mathbb{T}$

$$\begin{cases} {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{S}(t)) = \Pi^* - \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - \varphi^*\mathbb{S}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{E}(t)) = \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - (\varphi^* + \lambda^*)\mathbb{E} + (1 - \varrho^*)\delta^*\mathbb{T}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{I}(t)) = \lambda^*\mathbb{E} - (\varphi^* + \eta_1^* + \gamma^*)\mathbb{I}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{T}(t)) = \gamma^*\mathbb{I} - (\varphi^* + \eta_2^* + \delta^*)\mathbb{T}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{R}(t)) = \varrho^*\delta^*\mathbb{T} - \varphi^*\mathbb{R}, \\ \mathbb{S}(0) = \mathbb{S}_0, \quad \mathbb{E}(0) = \mathbb{E}_0, \quad \mathbb{I}(0) = \mathbb{I}_0, \quad \mathbb{T}(0) = \mathbb{T}_0, \quad \mathbb{R}(0) = \mathbb{R}_0. \end{cases} \quad (2)$$

Our paper is organized as: Section 1 is consisting of some literature. In Section 2, needful basic results are given. Some required materials are developed in 3. Theoretical approach is applied in Section 4. Numerical interpretation are given in Section 5. Brief concluding remarks are provided in 6.

2. Fundamental results

Definition 2.1. [52] The ABC derivative with FO $\ell \in [0, 1]$ for $g(t) \in \mathbb{H}^1[0, T]$ is given as

$${}^{\text{ABC}}\mathbb{D}_t^\ell(g(t)) = \frac{{}^{\text{ABC}}(\ell)}{1 - \ell} \int_0^t \mathbf{E}_\ell \left[\frac{-\ell}{1 - \ell} (t - z) \right] \frac{d}{dz} g(z) dz, \quad (3)$$

where $\text{ABC}(\ell) = \frac{\ell}{2 - \ell}$ is the normalization function and $\text{ABC}(0) = \text{ABC}(1) = 1$ and $\mathbf{E}_\ell$ is the “Mittag–Leffler function” which is as

$$\mathbf{E}_\ell(y) = \sum_{k=0}^{\infty} \frac{y^k}{\Gamma(\ell k + 1)}.$$

Definition 2.2. [52] Let $0 \leq \ell \leq 1$ and $g(t) \in \mathbb{L}^1[0, T]$, then the fractional integral in ABC sense is given by

$${}^{\text{ABC}}\mathbb{I}_0^\ell(g(t)) = \frac{1 - \ell}{{}^{\text{ABC}}(\ell)} g(t) + \frac{\ell}{{}^{\text{ABC}}(\ell)\Gamma(\ell)} \int_0^t (t - z)^{\ell-1} g(z) dz.$$

Lemma 2.3. [52] Corresponding to the problem

$$\begin{cases} {}^{\text{ABC}}\mathbb{D}_0^\ell g(t) = \Omega(t), \\ \Omega(0) = \Omega_0, \end{cases} \quad (4)$$

we get its solution in (5) as

$$g(t) = g_0 + \frac{1 - \ell}{{}^{\text{ABC}}(\ell)} g(t) + \frac{\ell}{{}^{\text{ABC}}(\ell)\Gamma(\ell)} \int_0^t (t - z)^{\ell-1} g(z) dz. \quad (5)$$

Theorem 2.4. [66] If closed and convex set $\mathcal{M} \neq \emptyset$ be subset of Banach space $\mathcal{B}$, and let ψ_1 and ψ_2 be the map from $\mathcal{M}$ to $\mathcal{B}$, which satisfy some conditions:

1. For all $\mu_1, \mu_2 \in \mathcal{M}$, then $\psi_1\mu_1 + \psi_2\mu_2 \in \mathcal{M}$;
2. ξ_1 is relatively compact and continuous;
3. Let $\chi > 0 \ni \|\psi_2\delta_1 - \psi_1\delta_2\| \leq \chi\|\delta_1 - \delta_2\|$, for all $\delta_1, \delta_2 \in \mathcal{B}$, then one has $\Omega \in \mathcal{B}$ with $\psi_1\Omega + \psi_2\Omega = \Omega$.

3. Computation of R_0 for model (2)

Here in this section, we provide the equilibrium points and $\mathfrak{R}_0$ for the considered model. Without $\mathbb{R}$ class we get the first four

equations of the model (2). The obtained new model without $\mathbb{R}$ for DFE (Disease-free equilibrium) and the $\mathfrak{R}_0$ is considered. The new model without recovered class is given along with the initial condition as:

$$\begin{cases} {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{S}(t)) = \Pi^* - \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - \varphi^*\mathbb{S}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{E}(t)) = \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - (\varphi^* + \lambda^*)\mathbb{E} + (1 - \varrho^*)\delta^*\mathbb{T}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{I}(t)) = \lambda^*\mathbb{E} - (\varphi^* + \eta_1^* + \gamma^*)\mathbb{I}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell(\mathbb{T}(t)) = \gamma^*\mathbb{I} - (\varphi^* + \eta_2^* + \delta^*)\mathbb{T}. \end{cases} \quad (6)$$

The possible region for the model (6) is below as:

$$\Theta^* = \left(\mathbb{S}(t), \mathbb{E}(t), \mathbb{I}(t), \mathbb{T}(t), \mathbb{R}(t) \in \mathbb{R}_+^5 \mid 0 \leq \frac{\Pi^*}{\varphi^*}, 0 \leq \mathbb{S}(t), \right. \\ \left. 0 \leq \mathbb{E}(t), 0 \leq \mathbb{I}(t), 0 \leq \mathbb{T}(t) \right).$$

For obtaining the DFE points, equating to zero, the right sides of (6), on has $\mathcal{E}^0 = (\mathbb{S}^0, 0, 0, 0)$, where

$$\mathbb{S}^0 = \frac{\Pi^*}{\varphi^*}.$$

Now, we examine $\mathfrak{R}_0$ for the model (6). To determine $\mathfrak{R}_0$, we use the next generation operator method [65]

$$\mathfrak{R}_0 = \frac{\lambda^* \xi^* \mathbb{S}^0 (\varphi^* + \eta_2^* + \delta^* + \varsigma^* \gamma^*)}{(\varphi^* + \lambda^*)(\varphi^* + \eta_1^* + \gamma^*)(\varphi^* + \eta_2^* + \delta^*) - (1 - \varrho^*)\delta^* \lambda^* \gamma^*}. \quad (7)$$

If $\varrho^* = 1$, then there is no incomplete treatment. If we change $\mathfrak{R}_0$ with $\mathfrak{R}_1$ then

$$\mathfrak{R}_1 = \frac{\lambda^* \xi^* \mathbb{S}^0 (\varphi^* + \eta_2^* + \delta^* + \varsigma^* \gamma^*)}{(\varphi^* + \lambda^*)(\varphi^* + \eta_1^* + \gamma^*)(\varphi^* + \eta_2^* + \delta^*)}, \quad (8)$$

clearly $\mathfrak{R}_1 < \mathfrak{R}_0$, thus incomplete treatment cause increase or decrease in TB infection.

For (6), we present the following theorem about local and global stability of DFE of the system (6)(detail is in [10]).

Theorem 3.1. Assume that $\mathfrak{R}_0 < 1$, the DFE $\mathcal{E}^0$ corresponding to (6) is stable locally. Otherwise unstable.

Theorem 3.2. Assume that $\mathfrak{R}_0 < 1$, the DFE point in Θ^* of the model (6) is stable globally. Otherwise unstable.

4. Fixed point approach

It is of tremendous attention to inquire whether a dynamical system we examine exists or not. Fixed point theory gives some detail about this need. We derive the required results for (2):

$$\begin{cases} {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{S}(t) = \mathbb{U}_1(\mathbb{S}(t), t) = \Pi^* - \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - \varphi^*\mathbb{S}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{E}(t) = \mathbb{U}_2(\mathbb{E}(t), t) = \xi^*\mathbb{S}(\mathbb{I} + \varsigma^*\mathbb{T}) - (\varphi^* + \lambda^*)\mathbb{E} + (1 - \varrho^*)\delta^*\mathbb{T}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{I}(t) = \mathbb{U}_3(\mathbb{I}(t), t) = \lambda^*\mathbb{E} - (\varphi^* + \eta_1^* + \gamma^*)\mathbb{I}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{T}(t) = \mathbb{U}_4(\mathbb{T}(t), t) = \gamma^*\mathbb{I} - (\varphi^* + \eta_2^* + \delta^*)\mathbb{T}, \\ {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{R}(t) = \mathbb{U}_5(\mathbb{R}(t), t) = \varrho^*\delta^*\mathbb{T} - \varphi^*\mathbb{R}. \end{cases} \quad (9)$$

For simplicity, we consider the proposed model as in (10)

$$\begin{aligned} {}^{\text{ABC}}\mathbb{D}_t^\ell \mathbb{G}(t) &= \mathbb{G}(t, \Omega(t)), \\ \Omega(0) &= \Omega_0, \end{aligned}$$

where

$$\Omega(t) = \begin{cases} \mathbb{S}(t) \\ \mathbb{E}(t) \\ \mathbb{I}(t) \\ \mathbb{T}(t) \\ \mathbb{R}(t) \end{cases}, \quad \Omega_0(t) = \begin{cases} \mathbb{S}_0 \\ \mathbb{E}_0 \\ \mathbb{I}_0 \\ \mathbb{T}_0 \\ \mathbb{R}_0 \end{cases}, \quad \mathbb{G}(t, \Omega(t)) = \begin{cases} \mathbb{U}_1(\mathbb{S}(t), t) \\ \mathbb{U}_2(\mathbb{E}(t), t) \\ \mathbb{U}_3(\mathbb{I}(t), t) \\ \mathbb{U}_4(\mathbb{T}(t), t) \\ \mathbb{U}_5(\mathbb{R}(t), t). \end{cases} \quad (10)$$

Using ABC fractional integral with initial condition, we obtain

$$\begin{aligned} \Omega(t) &= \Omega_0 + \frac{1-\ell}{\text{ABC}(\ell)} \mathbb{G}(t, \Omega(t)) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \\ &\quad \times \int_0^t (t-y)^{\ell-1} \mathbb{G}(y, \Omega(y)) dy. \end{aligned} \quad (11)$$

Let the Banach space be $\mathcal{X} = [0, T]$ as $\mathbb{B} = \mathcal{C}(\mathcal{X}, \mathbb{R}^5)$ with norm as

$$\|\Omega\| = \sup_{t \in \mathcal{X}} \{\Omega(t) : \Omega \in \mathbb{B}\},$$

for every $\Omega \in \mathbb{B}$ and $t \in [0, T]$ then the function $\mathbb{G}(t, \Omega(t))$ obeys the give assertions:

(B1) There exist two constant such as γ_G and β_G

$$|\mathbb{G}(t, \Omega(t))| \leq \gamma_G |\Omega| + \beta_G.$$

(B2) $\exists$ a constant such as λ_G

$$|\mathbb{G}(t, \Omega_1(t)) - \mathbb{G}(t, \Omega_2(t))| \leq L_G |\Omega_1 - \Omega_2|.$$

Let's define some operator Υ_1 and Υ_2

$$\Upsilon_1 \Omega(t) = \Omega_0 + \frac{1-\ell}{\text{ABC}(\ell)} \mathbb{G}(t, \Omega(t)),$$

$$\Upsilon_2 \Omega(t) = \frac{\ell}{\text{ABC}(\ell)} \frac{1}{\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} \mathbb{G}(y, \Omega(y)) dy,$$

where $\Upsilon_1 + \Upsilon_2 = \mathbb{B}$.

Theorem 4.1. Suppose that the growth and Lipschitz conditions hold under the following conditions, then the Eq. (11) has at least one solution,

- $\frac{1-\ell}{\text{ABC}(\ell)} \lambda_G \leq 1.$
- $\Delta_1 = \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \beta_G \leq 1.$
- $\Delta_2 = \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \gamma_G \leq 1.$

Proof. Suppose $\mathfrak{B}_\tau = \{\Omega \in \mathcal{B} : \|\Omega\| \leq \tau\}$ be closed convex set. Firstly we will show that for each $\Omega_1, \Omega_2 \in \mathfrak{B}_\tau$, $\Upsilon_1 \Omega_1 + \Upsilon_2 \Omega_2 \in \mathfrak{B}_\tau$, using growth condition for this, we have

$$\begin{aligned} \|\Upsilon_1 \Omega_1 + \Upsilon_2 \Omega_2\| &\leq \sup_{t \in [0, T]} \left(|\Omega_0| + \frac{1-\ell}{\text{ABC}(\ell)} \mathbb{G}(t, \Omega(t)) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} |\mathbb{G}(y, \Omega(y))| dy \right), \\ &\leq \left(|\Omega_0| + \frac{1-\ell}{\text{ABC}(\ell)} (\gamma_G \|\Omega\| + \beta_G) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} (\gamma_G \|\Omega\| + \beta_G) dy \right), \\ &= |\Omega_0| + \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \beta_G + \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \gamma_G \tau, \\ &= \Delta_1 + \Delta_2 \tau \leq \tau. \end{aligned}$$

It goes to show that $\Upsilon_1 \Omega_1 + \Upsilon_2 \Omega_2 \in \mathfrak{B}_\tau$.

Next we present that Υ_1 is contraction. For each $\Omega_1, \Omega_2 \in \mathfrak{B}_\tau$, using Lipschitz condition, we obtain

$$\begin{aligned} \|\Upsilon_1 \Omega_1 + \Upsilon_2 \Omega_2\| &= \sup_{t \in [0, T]} \frac{1-\ell}{\text{ABC}(\ell)} |\mathbb{G}(t, \Omega_1(t)) - \mathbb{G}(t, \Omega_2(t))|, \\ &\leq \frac{1-\ell}{\text{ABC}(\ell)} \lambda_G \sup_{t \in [0, T]} |\Omega_1(t) - \Omega_2(t)|, \\ &\leq \frac{1-\ell}{\text{ABC}(\ell)} \lambda_G \|\Omega_1 - \Omega_2\|, \end{aligned}$$

where $\frac{1-\ell}{\text{ABC}(\ell)} \lambda_G < 1$. Thus, Υ_1 is a contraction.

Next to prove that Υ_2 is relatively compact. Let us suppose $\Omega \in \mathfrak{B}_\tau$,

$$\begin{aligned} \|\Upsilon_2 \Omega\| &\leq \sup_{t \in [0, T]} \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} |\mathbb{G}(y, \Omega(y))| dy, \\ &\leq \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} \sup_{t \in [0, T]} [\gamma_G |\Omega| + \beta_G] dy, \\ &\leq \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} [\gamma_G \|\Omega\| + \beta_G] dy, \\ &\leq \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} [\gamma_G \tau + \beta_G]. \end{aligned}$$

This shows, Υ_2 is uniformly bounded on $\mathfrak{B}_\tau$. By the end we show that, Υ_2 is equi-continuous. Let $\Omega \in \mathfrak{B}_\tau$ and $t_1, t_2 \in [0, T]$, such that $t_1 < t_2$, then

$$\begin{aligned} \|\Upsilon_1 \Omega_1(t_1) + \Upsilon_2 \Omega_2(t_2)\| &\leq \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_{t_1}^{t_2} (t_2-y)^{\ell-1} |\mathbb{G}(y, \Omega(y))| dy, \\ &\quad + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^{t_2} \left[(t_1-y)^{\ell-1} - (t_2-y)^{\ell-1} \right] |\mathbb{G}(y, \Omega(y))| dy, \\ &\leq \frac{2[\gamma_G \tau + \beta_G]}{\text{ABC}(\ell)\Gamma(\ell)} \left[(t_2 - t_1)^\ell \right]. \end{aligned}$$

So $\|\Upsilon_1 \Omega_1(t_1) + \Upsilon_2 \Omega_2(t_2)\| \rightarrow 0$ as $t_2 \rightarrow t_1$. So, relatively compactness of the operator Υ_2 is followed. Hence, (11) has at least one solution.

Theorem 4.2. The integral equ(11) has a unique solution, when the Lipschitz condition holds

$$\left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \lambda_G < 1.$$

Proof. Each $\Omega, \Omega^* \in \mathcal{B}$ and $t \in [0, T]$, then

$$\begin{aligned} \|\Upsilon_1 \Omega(t) + \Upsilon_2 \Omega^*(t)\| &\leq \max_{t \in [0, T]} \frac{1-\ell}{\text{ABC}(\ell)} |\mathbb{G}(t, \Omega(t)) - \mathbb{G}(t, \Omega^*(t))| \\ &\quad + \max_{t \in [0, T]} \frac{1-\ell}{\text{ABC}(\ell)} \int_0^t (t-y)^{\ell-1} |\mathbb{G}(t, \Omega(t)) - \mathbb{G}(t, \Omega^*(t))| dy, \\ &\leq \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \lambda_G \|\Omega - \Omega^*\|. \end{aligned}$$

Due to assumption $1 > \left(\frac{1-\ell}{\text{ABC}(\ell)} + \frac{T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \lambda_G$, the operator Υ is contraction. By theorem of Banach contraction the equ (11) has a unique solution. Thus the considered model has a unique solution.

Now we are going to discuss the stability of Ulam-Hyres (UH) for the considered model (2).

Definition 4.3. If there exist $\phi > 0$, then the model (2) is UH stable, for any $\tilde{\epsilon} > 0$ and $\Omega \in \mathcal{B}$

$$|\text{ABC} \mathbb{D}_t^\ell \tilde{\Omega}(t) - \mathbb{G}(t, \tilde{\Omega}(t))| \leq \tilde{\epsilon}, \quad (12)$$

and $\Omega \in \mathcal{B}$ satisfying system (2) along with initial condition $\Omega(0) = \tilde{\Omega}(0) = \tilde{\Omega}_0$, $\exists \|\tilde{\Omega} - \Omega\| \leq \phi \tilde{\epsilon}$. where

$$\begin{cases} \tilde{\Omega} = (\tilde{S}, \tilde{E}, \tilde{I}, \tilde{T}, \tilde{R})^T, \\ \tilde{\Omega}_0 = (\tilde{S}_0, \tilde{E}_0, \tilde{I}_0, \tilde{T}_0, \tilde{R}_0)^T, \\ \mathbb{G}(t, \tilde{\Omega}(t)) = (\tilde{U}_1, \tilde{U}_2, \tilde{U}_3, \tilde{U}_4, \tilde{U}_5)^T, \\ \tilde{\epsilon} = \max(\epsilon_j)^T; \quad j = 1, 2, \dots, 5, \\ \phi = \max(\phi_j)^T; \quad j = 1, 2, \dots, 5. \end{cases} \quad (13)$$

Remark 4.4. Considering a small perturbation such as $\Phi \in \mathbb{C}[0, T]$ and $\Phi(0) = 0$ with some solution as:

- $|\Phi(t)| \leq \epsilon$ and $\epsilon > 0$.
- ${}^{\text{ABC}}\mathbb{D}_t^\ell \tilde{\Omega}(t) = \mathbb{G}(t, \tilde{\Omega}(t)) + \Phi(t)$.

Lemma 4.5. The perturbed problem solution is as

$$\begin{aligned} {}^{\text{ABC}}\mathbb{D}_t^\ell \tilde{\Omega}(t) &= \mathbb{G}(t, \tilde{\Omega}(t)) + \Phi(t), \\ \Omega(0) &= \Omega_0, \end{aligned} \quad (14)$$

which satisfies

$$|\tilde{\Omega}_\Phi(t) - \tilde{\Omega}(t)| \leq \left(\frac{(1-\ell)\Gamma(\ell) + T^\ell}{\text{ABC}(\ell)\Gamma(\ell)} \right) \leq \epsilon. \quad (15)$$

Proof. The approximate solution of (14) is identical to the given problem

$$\begin{aligned} \tilde{\Omega}_\Phi(t) &= \tilde{\Omega}_0 + \frac{1-\ell}{\text{ABC}(\ell)} \mathbb{G}(t, \tilde{\Omega}(t)) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} \mathbb{G}(y, \tilde{\Omega}(y)) dy \\ &\quad + \frac{1-\ell}{\text{ABC}(\ell)} \Phi(t) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} \Phi(y) dy, \end{aligned} \quad (16)$$

and

$$\tilde{\Omega}(t) = \tilde{\Omega}_0 + \frac{1-\ell}{\text{ABC}(\ell)} \mathbb{G}(t, \tilde{\Omega}(t)) + \frac{\ell}{\text{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} \mathbb{G}(y, \tilde{\Omega}(y)) dy. \quad (17)$$

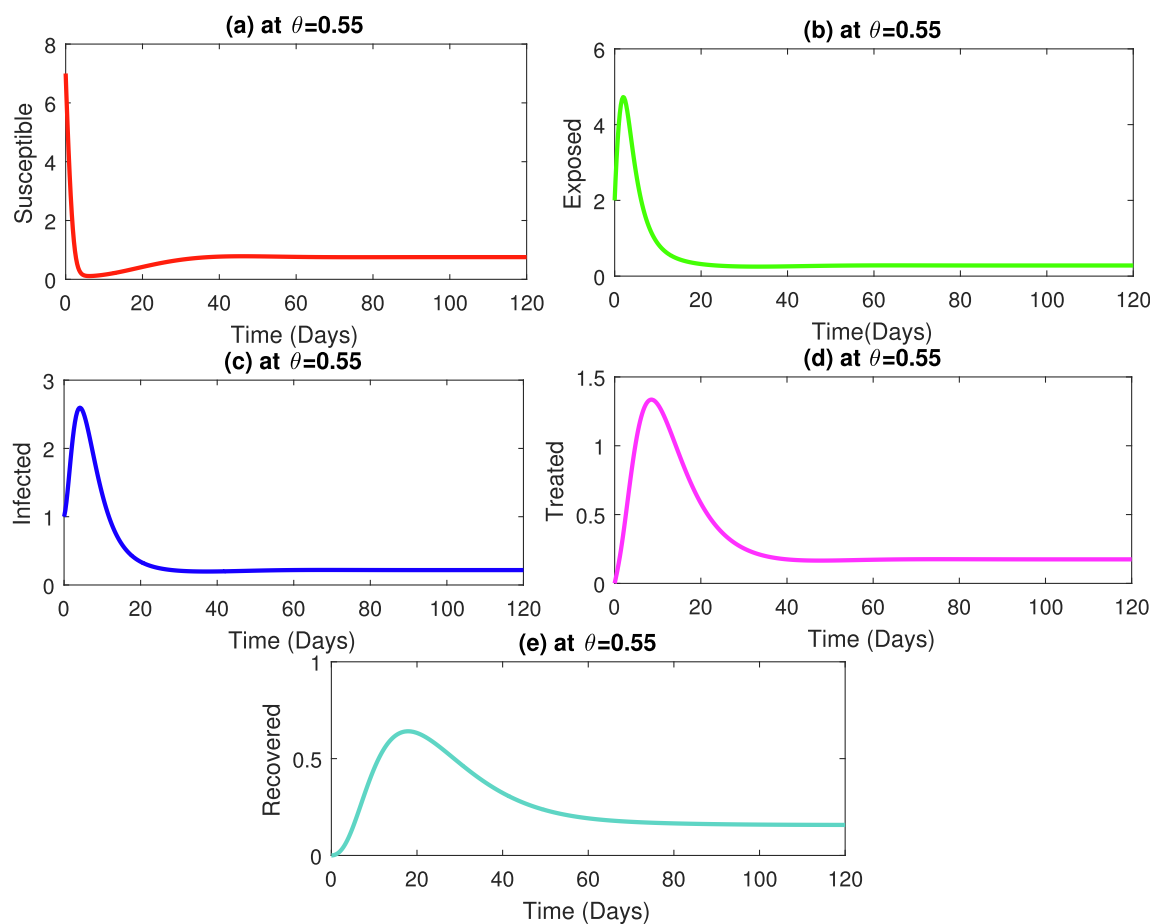


Fig. 1 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 0.55$, showing convergency and stability.

Using remark (1), we have

$$\begin{aligned} |\tilde{\Omega}_\Phi(t) - \tilde{\Omega}(t)| &\leq \frac{1-\ell}{\mathbb{ABC}(\ell)} |\Phi(t)| + \frac{\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} |\Phi(y)| dy, \\ &\leq \left(\frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \right) \leq \epsilon. \end{aligned} \quad (18)$$

Theorem 4.6. Under the assumption **B2** the system (2) is UH stable.

Proof. Let $\Omega \in \mathcal{B}$ be the solution for (12), as uniqueness $\Omega \in \mathcal{B}$ be the atmost one solution for model (2) along with $\Omega(0) = \tilde{\Omega}_0$ and

$$\begin{aligned} \Omega(t) &= \tilde{\Omega}_0 + \frac{1-\ell}{\mathbb{ABC}(\ell)} \mathbb{G}(t, \Omega(t)) + \frac{\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \\ &\quad \times \int_0^t (t-y)^{\ell-1} \mathbb{G}(y, \Omega(y)) dy, \end{aligned}$$

using the above lemma, we get

$$\begin{aligned} |\tilde{\Omega}_\Phi(t) - \Omega(t)| &\leq |\tilde{\Omega}_\Phi(t) - \tilde{\Omega}(t)| + |\tilde{\Omega}_\Phi(t) - \Omega(t)|, \\ &\leq 2 \left(\frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \right) \epsilon + \frac{1-\ell}{\mathbb{ABC}(\ell)} |\mathbb{G}(t, \tilde{\Omega}(t)) - \mathbb{G}(t, \Omega(t))| \\ &\quad + \frac{\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \int_0^t (t-y)^{\ell-1} |\mathbb{G}(y, \tilde{\Omega}(y)) - \mathbb{G}(y, \Omega(y))| dy, \\ &\leq 2 \left(\frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \right) \epsilon + \left(\frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \right) \lambda_G \|\tilde{\Omega} - \Omega\|, \end{aligned}$$

which means that

$$\|\tilde{\Omega} - \Omega\| \leq \frac{2\Lambda\epsilon}{1-\alpha}, \quad (19)$$

where

$$\begin{aligned} \Lambda &= \frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)}, \\ \alpha &= \left(\frac{(1-\ell)\Gamma(\ell)+T^\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \right) \lambda_G. \end{aligned} \quad (20)$$

For $\phi = \frac{2\Lambda}{1-\alpha}$. So Eq. (19) can be written as $\|\tilde{\Omega} - \Omega\| \leq \phi\epsilon$. Thus, the considered model (2) is UH stable.

5. Numerical scheme for model (2) by ABC

We proceed to obtain the numerical results for our proposed model (2). We applied the modified Adams–Bashforth technique to assume the numerical procedure for the simulation as used in [57,66]. For this purpose, we applied the fractional AB integral operator to get a numerical simulation on behalf of our model (2). Consider the model (2) as,

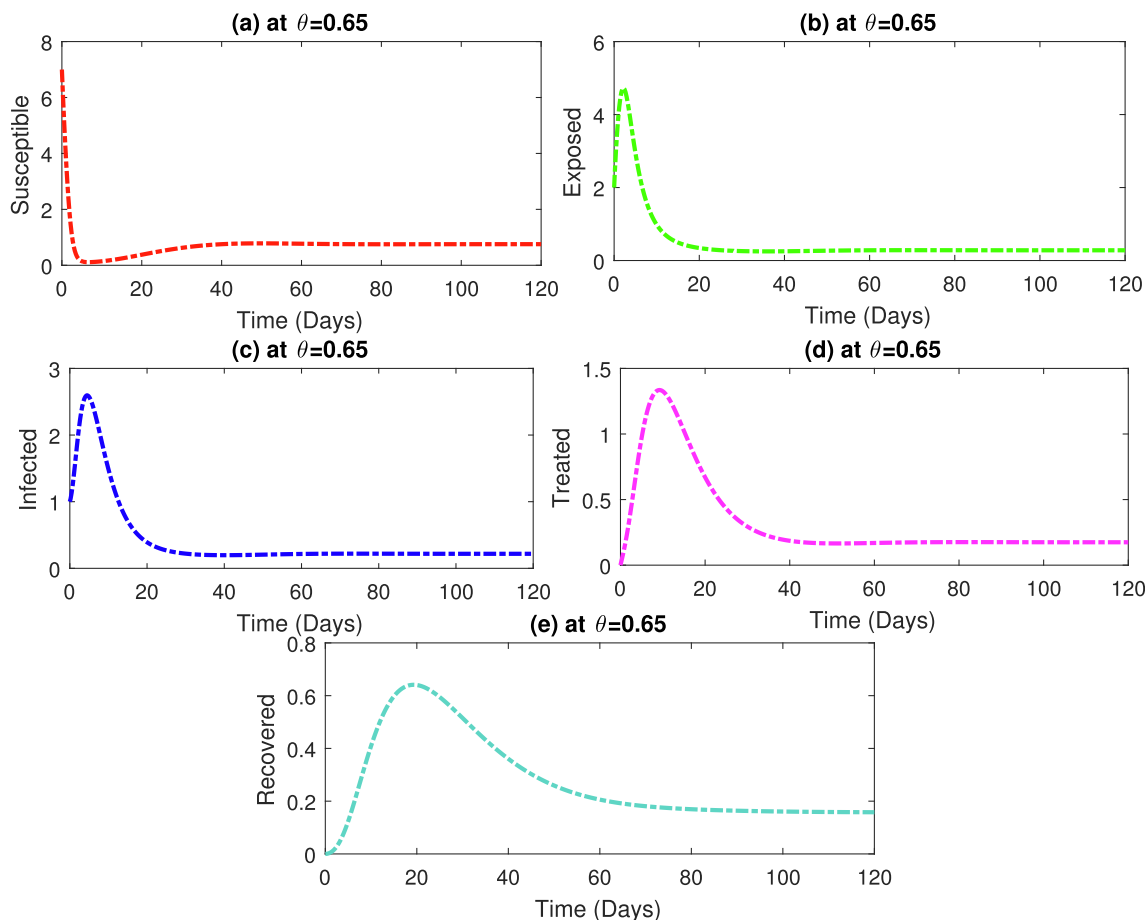


Fig. 2 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 0.65$, showing convergency and stability.

$$\begin{cases} {}^{ABC}D_t^\ell S(t) = G_1(S(t), t) = \Pi^* - \zeta^* S(\mathbb{I} + \zeta^* \mathbb{T}) - \varphi^* S, \\ {}^{ABC}D_t^\ell E(t) = G_2(E(t), t) = \zeta^* S(\mathbb{I} + \zeta^* \mathbb{T}) - (\varphi^* + \lambda^*) E + (1 - \varrho^*) \delta^* \mathbb{T}, \\ {}^{ABC}D_t^\ell I(t) = G_3(I(t), t) = \lambda^* E - (\varphi^* + \eta_1^* + \gamma^*) I, \\ {}^{ABC}D_t^\ell T(t) = G_4(T(t), t) = \gamma^* I - (\varphi^* + \eta_2^* + \delta^*) T, \\ {}^{ABC}D_t^\ell R(t) = G_5(R(t), t) = \varrho^* \delta^* T - \varphi^* R, \end{cases} \quad (21)$$

taking the fractional integral of Eq. (21) with the initial condition, we obtain

$$\begin{aligned} S(t) &= S(0) + \frac{(1-\ell)}{ABC(\ell)} [G_1(S(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^t (t-z)^{\ell-1} G_1(S(z), t) dz, \\ E(t) &= E(0) + \frac{(1-\ell)}{ABC(\ell)} [G_2(E(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^t (t-z)^{\ell-1} G_2(E(z), t) dz, \\ I(t) &= I(0) + \frac{(1-\ell)}{ABC(\ell)} [G_3(I(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^t (t-z)^{\ell-1} G_3(I(z), t) dz, \\ T(t) &= T(0) + \frac{(1-\ell)}{ABC(\ell)} [G_4(T(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^t (t-z)^{\ell-1} G_4(T(z), t) dz, \\ R(t) &= R(0) + \frac{(1-\ell)}{ABC(\ell)} [G_5(R(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^t (t-z)^{\ell-1} G_5(R(z), t) dz, \end{aligned} \quad (22)$$

considering the first equation from the system (22) to obtain an iterative scheme

$$\begin{aligned} S(t) &= S(0) + \frac{(1-\ell)}{ABC(\ell)} [G_1(S(t), t)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \\ &\quad \times \int_0^t (t-z)^{\ell-1} G_1(S(z), t) dz. \end{aligned}$$

Set $t = t_{\beta+1}$ for $r = 0, 1, 2, \dots$, the above equation becomes

$$\begin{aligned} S(t_{\beta+1}) &= S(0) + \frac{(1-\ell)}{ABC(\ell)} [G_1(S(t_\beta), t_\beta)] + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \int_0^{t_{\beta+1}} (t_{\beta+1}-z)^{\ell-1} G_1(S(z), t) dz, \\ &= S(0) + \frac{(1-\ell)}{ABC(\ell)} [G_1(S(t_\beta), t_\beta)] \\ &\quad + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \sum_{q=0}^r \int_{t_\beta}^{t_{\beta+1}} (t_{\beta+1}-z)^{\ell-1} G_1(S(z), t) dz. \end{aligned} \quad (23)$$

Now, using the term of interpolation polynomials we can address the interval $[t_\beta, t_{\beta+1}]$, the approximate function $G_1(S(t), t)$ as follows

$$\begin{aligned} G_1(S(t_\beta), t_\beta) &\cong \frac{G_1(S(t_\beta), t_\beta)}{h} (t - t_{\beta-1}) + \frac{G_1(S(t_{\beta-1}), t_{\beta-1})}{h} \\ &\quad \times (t - t_\beta), \end{aligned}$$

which implies that

$$\begin{aligned} S(t_{\beta+1}) &= S(0) + \frac{(1-\ell)}{ABC(\ell)} [G_1(S(t_\beta), t_\beta)] \\ &\quad + \frac{\ell}{ABC(\ell)\Gamma(\ell)} \sum_{q=0}^r \left(\frac{G_1(S(t_\beta), t_\beta)}{h} \int_{t_q}^{t_{q+1}} (t - t_{\beta-1})(t_{q+1} - t)^{\ell-1} dt \right. \\ &\quad \left. - \frac{G_1(S(t_{\beta-1}), t_{\beta-1})}{h} \int_{t_q}^{t_{q+1}} (t - t_q)(t_{\beta+1} - t)^{\ell-1} dt \right). \end{aligned} \quad (24)$$

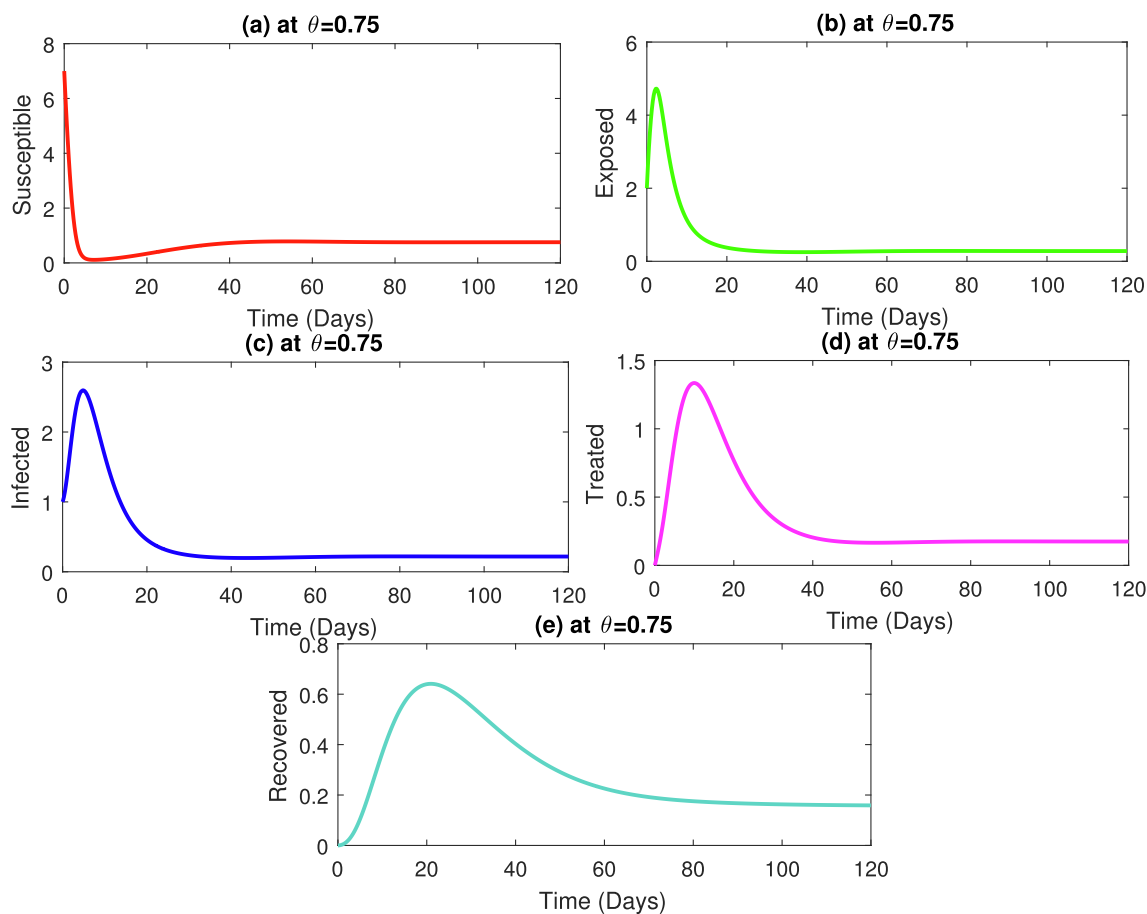


Fig. 3 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 0.75$, showing convergency and stability.

Let us calculate the integral as

$$\begin{aligned}\mathbb{I}_{q-1,\ell} &= \int_{t_q}^{t_{q+1}} (t - t_{q-1})(t_{\beta+1} - t)^{\ell-1} dt, \\ \mathbb{I}_{q,\ell} &= \int_{t_q}^{t_{q+1}} (t - t_q)(t_{\beta+1} - t)^{\ell-1} dt.\end{aligned}$$

So Eq. (25) becomes

$$\begin{aligned}\mathbb{S}(t) &= \mathbb{S}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_1(\mathbb{S}(t_\beta), t_\beta)] + \frac{\ell}{\mathbb{ABC}(\ell)\Gamma(\ell)} \sum_{q=0}^r \left(\frac{\mathbb{G}_1(\mathbb{S}(t_\beta), t_\beta)}{h} \mathbb{I}_{q-1,\ell} \right. \\ &\quad \left. - \frac{\mathbb{G}_1(\mathbb{S}(t_{\beta-1}), t_{\beta-1})}{h} \mathbb{I}_{q,\ell} \right).\end{aligned}\quad (25)$$

Next, we calculate the integrals $\mathbb{I}_{q-1,\ell}$ and $\mathbb{I}_{q,\ell}$ as follows

$$\begin{aligned}\mathbb{I}_{q-1,\ell} &= \int_{t_q}^{t_{q+1}} (t - t_{q-1})(t_{\beta+1} - t)^{\ell-1} dt \\ &= -\frac{1}{\ell} \left[(t_{q+1} - t_{q-1})(t_{\beta+1} - t_{q+1})^\ell - (t_q - t_{q-1})(t_{\beta+1} - t_q)^\ell \right] \\ &\quad - \frac{1}{\ell(\ell-1)} \left[(t_{\beta+1} - t_{q+1})^{\ell+1} - (t_{\beta+1} - t_q)^{\ell+1} \right],\end{aligned}$$

put $t_q = qh$, we get

$$\begin{aligned}\mathbb{I}_{q-1,\ell} &= -\frac{h^{\ell+1}}{\ell} \left[(q+1-(q-1))(\beta+1-(q+1))^\ell - (q-(q-1))(\beta+1-q)^\ell \right] \\ &\quad - \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta+1-(q+1))^{\ell+1} - (\beta+1-q)^{\ell+1} \right], \\ &= -\frac{h^{\ell+1}}{\ell(\ell-1)} \left[-2(\ell+1)(\beta-q)^\ell + (\ell+1)(\beta+1-q)^\ell - (\beta-q)^{\ell+1} + (\beta+1-q)^{\ell+1} \right], \\ &= \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta-q)^\ell (-2(\ell+1) - (\beta-q)) + (\beta+1-q)^\ell (\ell+1+\beta+1-q) \right], \\ &= \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta+1-q)^\ell (\beta-q+2+\ell) - (\beta-q)^\ell (\beta-q+2+2\ell) \right],\end{aligned}\quad (26)$$

and

$$\begin{aligned}\mathbb{I}_{q,\ell} &= \int_{t_q}^{t_{q+1}} (t - t_q)(t_{\beta+1} - t)^{\ell-1} dt, \\ &= -\frac{1}{\ell} \left[(t_{q+1} - t_q)(t_{\beta+1} - t_{q+1})^\ell \right], \\ &\quad - \frac{1}{\ell(\ell-1)} \left[(t_{\beta+1} - t_{q+1})^{\ell+1} - (t_{\beta+1} - t_q)^{\ell+1} \right],\end{aligned}$$

put $t_q = qh$, we have

$$\begin{aligned}\mathbb{I}_{q,\ell} &= -\frac{h^{\ell+1}}{\ell} \left[(q+1-q)(\beta+1-(q+1))^\ell \right] - \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta+1-(q+1))^{\ell+1} - (\beta+1-q)^{\ell+1} \right], \\ &= \frac{h^{\ell+1}}{\ell(\ell-1)} \left[-(\ell+1)(\beta-q)^\ell - (\beta-q)^{\ell+1} + (\beta+1-q)^{\ell+1} \right], \\ &= \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta-q)^\ell (-(\ell+1) - (\beta-q)) + (\beta+1-q)^{\ell+1} \right], \\ &= \frac{h^{\ell+1}}{\ell(\ell-1)} \left[(\beta+1-q)^{\ell+1} - (\beta-q)^\ell (\beta-q+1+\ell) \right].\end{aligned}\quad (27)$$

Substituting Eqs. (26) and (27) into (25), we get

$$\begin{aligned}\mathbb{S}(t_{\beta+1}) &= \mathbb{S}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_1(\mathbb{S}(t_\beta), t_\beta)] + \frac{\ell}{\mathbb{ABC}(\ell)} \sum_{q=0}^{\beta} \left(\frac{\mathbb{G}_1(\mathbb{S}(t_\beta), t_\beta)}{\Gamma(\ell+2)} \times \right. \\ &\quad \left. h^\ell \left[(\beta+1-q)^\ell (\beta-q+2+\ell) - (\beta-q)^\ell (\beta-q+2+2\ell) \right] - \frac{\mathbb{G}_1(\mathbb{S}(t_{\beta-1}), t_{\beta-1})}{\Gamma(\ell+2)} \right. \\ &\quad \left. \times h^\ell \left[(\beta+1-q)^{\ell+1} - (\beta-q)^\ell (\beta-q+1+\ell) \right] \right).\end{aligned}\quad (28)$$

Also for the rest of the equations of the model, as

$$\begin{aligned}\mathbb{E}(t_{\beta+1}) &= \mathbb{E}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_2(\mathbb{E}(t_\beta), t_\beta)] + \frac{\ell}{\mathbb{ABC}(\ell)} \sum_{q=0}^{\beta} \left(\frac{\mathbb{G}_2(\mathbb{E}(t_\beta), t_\beta)}{\Gamma(\ell+2)} \times \right. \\ &\quad \left. h^\ell \left[(\beta+1-q)^\ell (\beta-q+2+\ell) - (\beta-q)^\ell (\beta-q+2+2\ell) \right] - \frac{\mathbb{G}_2(\mathbb{E}(t_{\beta-1}), t_{\beta-1})}{\Gamma(\ell+2)} \right. \\ &\quad \left. \times h^\ell \left[(\beta+1-q)^{\ell+1} - (\beta-q)^\ell (\beta-q+1+\ell) \right] \right),\end{aligned}\quad (29)$$

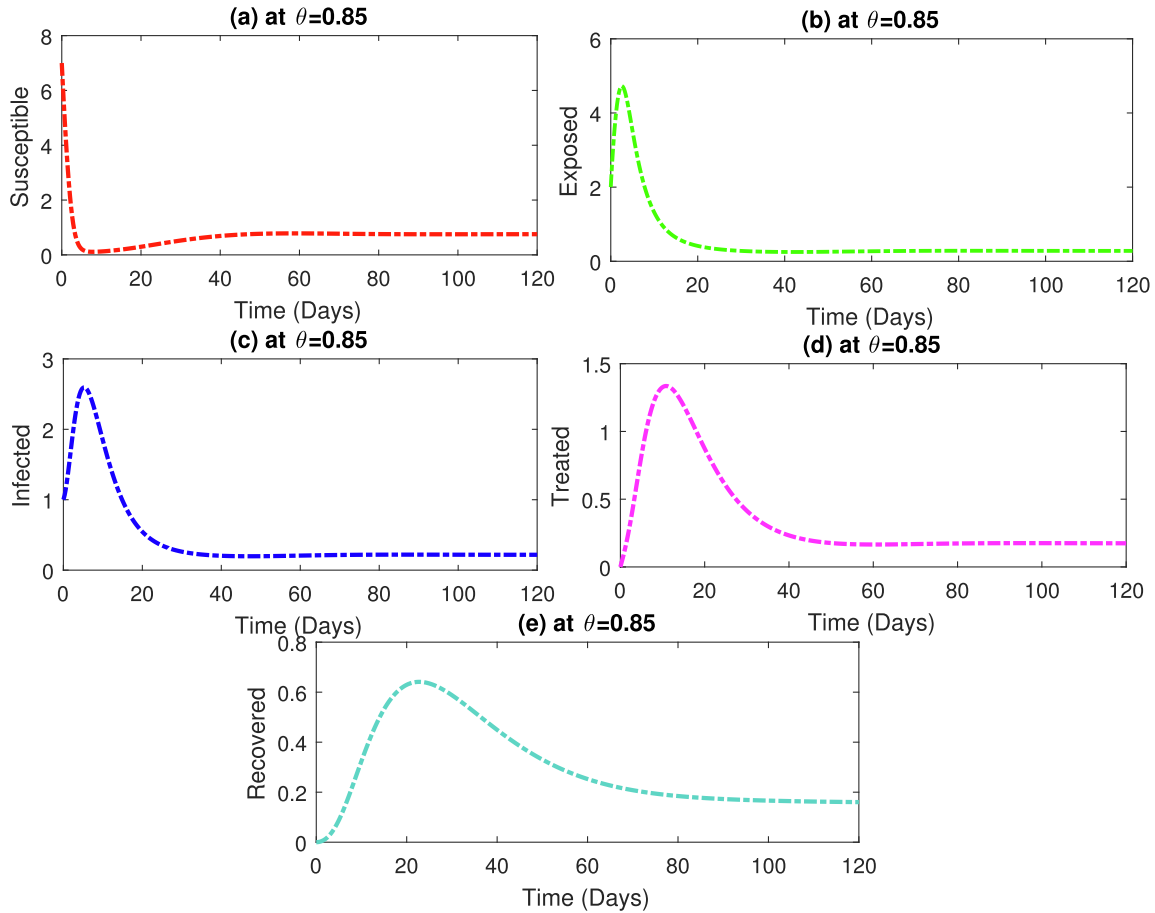


Fig. 4 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 0.85$, showing convergence and stability.

$$\begin{aligned} \mathbb{I}(t_{\beta+1}) = & \mathbb{I}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_3(\mathbb{S}(t_{\beta}), t_{\beta})] + \frac{\ell}{\mathbb{ABC}(\ell)} \sum_{q=0}^{\beta} \left(\frac{\mathbb{G}_3(\mathbb{I}(t_{\beta}), t_{\beta})}{\Gamma(\ell+2)} \times \right. \\ & h^{\ell} [(\beta+1-q)^{\ell} (\beta-q+2+\ell) - (\beta-q)^{\ell} (\beta-q+2+2\ell)] - \frac{\mathbb{G}_3(\mathbb{I}(t_{\beta-1}), t_{\beta-1})}{\Gamma(\ell+2)} \\ & \left. \times h^{\ell} [(\beta+1-q)^{\ell+1} - (\beta-q)^{\ell} (\beta-q+1+\ell)] \right), \end{aligned} \quad (30)$$

$$\begin{aligned} \mathbb{T}(t_{\beta+1}) = & \mathbb{T}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_4(\mathbb{T}(t_{\beta}), t_{\beta})] + \frac{\ell}{\mathbb{ABC}(\ell)} \sum_{q=0}^{\beta} \left(\frac{\mathbb{G}_4(\mathbb{T}(t_{\beta}), t_{\beta})}{\Gamma(\ell+2)} \times \right. \\ & h^{\ell} [(\beta+1-q)^{\ell} (\beta-q+2+\ell) - (\beta-q)^{\ell} (\beta-q+2+2\ell)] - \frac{\mathbb{G}_4(\mathbb{T}(t_{\beta-1}), t_{\beta-1})}{\Gamma(\ell+2)} \\ & \left. \times h^{\ell} [(\beta+1-q)^{\ell+1} - (\beta-q)^{\ell} (\beta-q+1+\ell)] \right), \end{aligned} \quad (31)$$

$$\begin{aligned} \mathbb{R}(t_{\beta+1}) = & \mathbb{R}(0) + \frac{(1-\ell)}{\mathbb{ABC}(\ell)} [\mathbb{G}_5(\mathbb{R}(t_{\beta}), t_{\beta})] + \frac{\ell}{\mathbb{ABC}(\ell)} \sum_{q=0}^{\beta} \left(\frac{\mathbb{G}_5(\mathbb{R}(t_{\beta}), t_{\beta})}{\Gamma(\ell+2)} \times \right. \\ & h^{\ell} [(\beta+1-q)^{\ell} (\beta-q+2+\ell) - (\beta-q)^{\ell} (\beta-q+2+2\ell)] - \frac{\mathbb{G}_5(\mathbb{R}(t_{\beta-1}), t_{\beta-1})}{\Gamma(\ell+2)} \\ & \left. \times h^{\ell} [(\beta+1-q)^{\ell+1} - (\beta-q)^{\ell} (\beta-q+1+\ell)] \right). \end{aligned} \quad (32)$$

5.1. Numerical procedure

In this section, we have to find the approximate solution of the proposed system (2). For this, the different parameter values are considered as given in Table 2. Using the parameter values

given in the following table 2, we simulate the compartment of the considered model from Figs. 1–6 for different fractional or arbitrary order. We observe from all the six figures that fractional-order values are comparable with integer-order values and show the validity of the fractional analysis of the model. From Figs. 1–6 the susceptible cases have been shown and behave rapidly decreasing in it up to two weeks. After two or three-week the susceptible class shows a small growth and then becomes convergent and stable. The rapid decrease is more at smaller orders and smaller at higher orders. Similarly, Figs. 1–6 the class of exposed population grows for a week at various non-integer order of derivatives. With time for two or three weeks, it decays and becomes stable. At the fractional orders, all the compartments behave like in integer-order at the points of stability. The infected cases showed in Figs. 1–6 increases at the initial stages and raises to a maximum peak value of 2.12 at various non-integer order as no cure and precaution have applied. When the treatment and precaution are utilized then the infected people are decreased to 0.35 and showing that the fractional-order values tend to integer order. As the infected class increased the cure and treatment also necessary to be maximized at the initial stages on the different fractional-order of a derivative. Check-up and hospitalization will finish the TB from the different territories and as the infection will stable and convergent the treated class becomes stable. We have finalized that recover cases in Figs. 1–6 grows

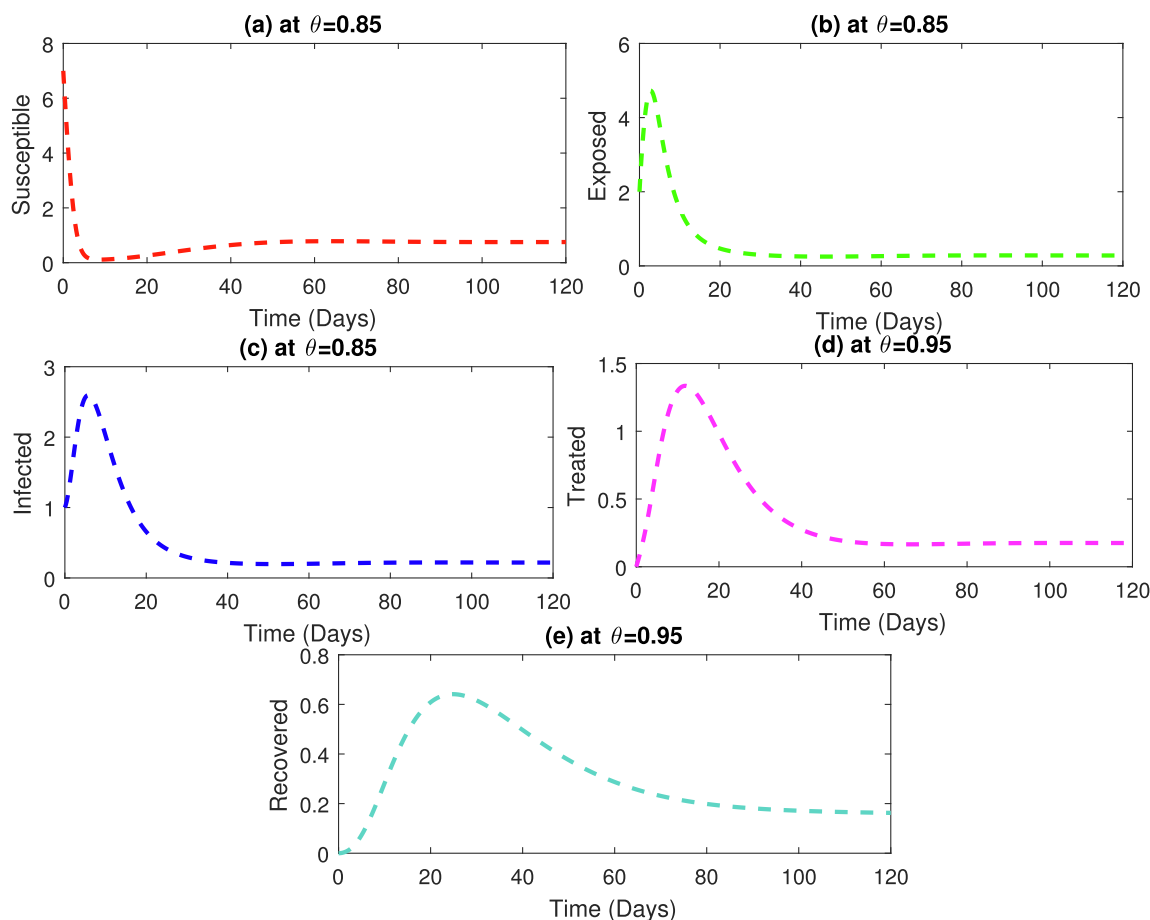


Fig. 5 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 0.95$, showing convergency and stability.

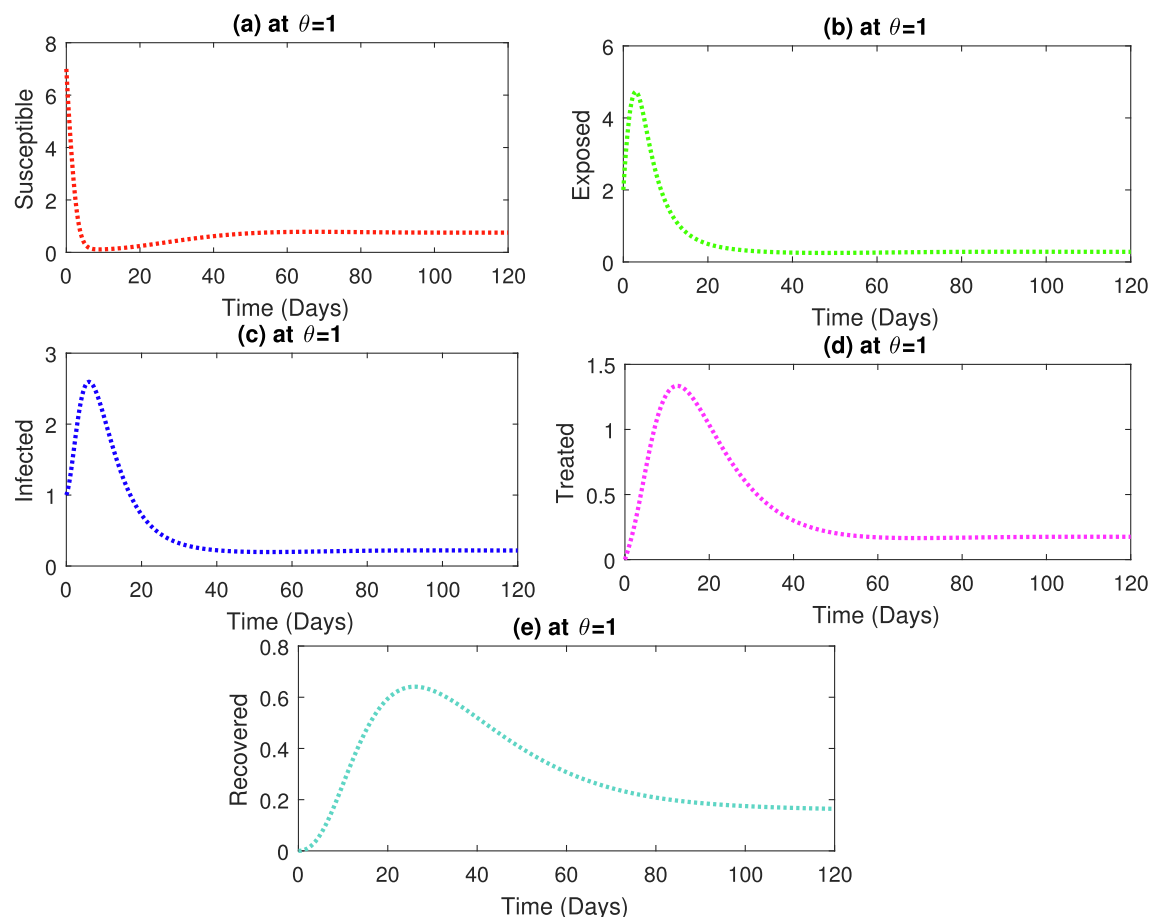


Fig. 6 Plots of the numerical results of the proposed model (2) for different compartments at $\theta = \ell = 1$, showing convergency and stability.

to 0.63 at the very beginning on different fractional-order and by treatment and checkup, the recovered class become stable and convergent.

6. Concluding remarks

In this paper, we have examined the FO model of TB under the effective contact rate, treatment rate and incomplete treatment against efficient treatment under the ABC derivative approach. We have also provided the equilibrium point and the stability results by applying the techniques and theorems of the basic reproduction number. Through fixed point approach some existence and stable results have been studied. Further, some numerical results for the proposed model have developed by AB method. The numerical results have been interpreted through graphically for various FO. The graphical representation provides a research gape to the scholars that mathematical models can be much more significantly analyzed by fractional-order as compared to integer-order by providing the information between two numbers. All the compartments simulations of the FO model have shown stability and convergency.

Declaration of Competing Interest

None.

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References

- [1] World Health Organization. Tackling the dual burden of TB and diabetes for patients and their families, 14 November, 2019.
- [2] S.E. Geerlings, A.I. Hoepelman, Immune dysfunction in patients with diabetes mellitus (DM), *FEMS Immunol. Med. Microbiol.* 26 (3–4) (1999) 259–265.
- [3] H. Waaler, A. Geser, S. Andersen, The use of mathematical models in the study of the epidemiology of tuberculosis, *Am. J. Public Health Nations Health* 52 (1962) 1002–1013.
- [4] Y. Yang, J. Li, Z. Ma, L. Liu, Global stability of two models with incomplete treatment for tuberculosis, *Chaos, Solitons Fractals* 43 (2010), 79–85.
- [5] J. Liu, T. Zhang, Global stability for a tuberculosis model, *Math. Comp. Model.* 54 (2011) 836–845.
- [6] D. Okuonghae, A mathematical model of tuberculosis transmission with heterogeneity in disease susceptibility and

- progression under a treatment regime for infectious cases, *Appl. Math. Model* 37 (2013) 6786–6808.
- [7] J. Zhang, Y. Liand, X. Zhang, Mathematical modeling of tuberculosis data of china, *J. Theor. Bio.* 365 (2015) 159–163.
 - [8] S.R. Wallis, D. Okuonghae, Mathematical models of tuberculosis reactivation and relapse, *Front. Microbiol.* 7 (2016) 1–7.
 - [9] A.O. Egonmwan, D. Okuonghae, Analysis of a mathematical models for tuberculosis with diagnosis, *J. Appl. Math. Comput.* 59 (1–2) (2019) 129–162.
 - [10] I. Ullah, S. Ahmad, Q. Al-Mdallal, Z.A. Khan, H. Khan, A. Khan, Stability analysis of a dynamical model of tuberculosis with incomplete treatment, *Adv. Diff. Eqs.* 2020 (1) (2020) 1–14.
 - [11] J. Danane, K. Allali, Z. Hammouch, Mathematical analysis of a fractional differential model of HBV infection with antibody immune response, *Chaos, Solitons & Fractals* 136 (2020) 109787.
 - [12] W. Gao, P. Veerasha, H.M. Baskonus, D.G. Prakasha, P. Kumar, A New Study of Unreported Cases of 2019-nCoV Epidemic Outbreaks, *Chaos, Solitons & Fractals* 138 (2020) 109929.
 - [13] M. Goyal, H.M. Baskonus, A. Prakash, Regarding new positive, bounded and convergent numerical solution of nonlinear time fractional HIV/AIDS transmission model, *Chaos, Solitons & Fractals* 139 (2020) 110096.
 - [14] S. Ullah, M.A. Khan, M. Farooq, Z. Hammouch, D. Baleanu, A Fractional Model for the Dynamics of Tuberculosis Infection Using Caputo-Fabrizio Derivative, *Discrete & Continuous Dynamical Systems-S*, 13(3)(2020) 975.
 - [15] M. Goyal, H.M. Baskonus, A. Prakash, An efficient technique for a time fractional model of lassa hemorrhagic fever spreading in pregnant women, *Eur. Phys. J. Plus* 134 (2019) 1–10.
 - [16] W. Gao, P. Veerasha, D.G. Prakasha, H.M. Baskonus, G. Yel, New approach for the model describing the deathly disease in pregnant women using Mittag-Leffler function, *Chaos, Solitons and Fractals* 134 (2020) 109696.
 - [17] D. Kumar, J. Singh, M. Al-Qurashi, D. Baleanu, A new fractional SIRS-SI malaria disease model with application of vaccines, anti-malarial drugs, and spraying, *Adv. Diff Eqs.* 278 (2019) 1–10.
 - [18] K. Shah, M.A. Alqudah, F. Jarad, T. Abdeljawad, Semi-analytical study of Pine Wilt disease model with convex rate under Caputo-Fabrizio fractional order derivative, *Chaos, Solitons & Fractals* 135 (2020) 109754.
 - [19] I. Ullah, S. Ahmad, M. Ur Rahman, M. Arfan, Investigation of fractional order tuberculosis (TB) model via Caputo derivative, *Chaos, Solitons & Fractals* 138 (2020) 110479.
 - [20] R. Toledo-Hernandez, V. Rico-Ramirez, A. Gustavo, Iglesias-Silva, U.N. Diwekar, A fractional calculus approach to the dynamic optimization of biological reactive systems. part i: fractional models for biological reactions, *Chem. Eng. Sci.*, 117 (2014) 217–228.
 - [21] K.S. Miller, B. Ross, An introduction to the fractional calculus and fractional differential equations, Wiley, New York, 1993.
 - [22] A.A. Kilbas, H.M. Srivastava, J.J. Trujillo, Theory and Application of Fractional Differential Equations, Elsevier, Amester Dam, North Holland, 204, (2006).
 - [23] Fatmawati, M.A. Khan, E. Bonyah, Z. Hammouch, E.M. Shaiful, A mathematical model of tuberculosis (TB) transmission with children and adults groups: A fractional model, *AIMS: Math.* 5(4) (2020) 2813–2842.
 - [24] W. Gao, P. Veerasha, D.G. Prakasha, H.M. Baskonus, G. Yel, New numerical results for the time-fractional Phi-four equation using a novel analytical approach, *Symmetry* 12 (3) (2020) 478.
 - [25] W. Gao, P. Veerasha, D.G. Prakasha, H.M. Baskonus, New numerical simulation for fractional Benney-Lin equation arising in falling film problems using two novel techniques, *Numer. Methods Partial Diff. Eqs.* 37 (1) (2020) 210–243.
 - [26] D. Baleanu, K. Diethelm, E. Scalas, J.J. Trujillo, Fractional Calculus: Models and Numerical Methods, World Scientific, Singapor, 2012.
 - [27] Y.A. Rossikhin, M.V. Shitikova, Applications of fractional calculus to dynamic problems of linear and nonlinear hereditary mechanics of solids, *Appl. Mech. Rev.* 50 (1997) 15–67.
 - [28] V. Lakshmikantham, S. Leela, J.V. Devi, Theory of Fractional Dynamic Systems, CSP (2009).
 - [29] A. Khan, K. Shah, M. Arfan, T. Abdeljawad, F. Jarad, A New Numerical Treatment for Fractional Differential Equations Based on non Discretization of data using Laguerre polynomials, *Fractals* 28 (8) (2020) 2040046.
 - [30] H. Khan, Y. Li, A. Khan, A. Khan, Existence of solution for a fractional order lotka-volterra reaction diffusion model with mittag-leffler kernel, *Math. Meth. Appl. Sci.* 42 (9) (2019) 3377–3387.
 - [31] R.A. Khan, K. Shah, Existence and uniqueness of solutions to fractional order multi-point boundary value problems, *Commun. Appl. Anal.* 19 (2015) 515–526.
 - [32] A.S. Shaikh, K.S. Nisar, Transmission dynamics of fractional order Typhoid fever model using Caputo-Fabrizio operator, *Chaos, Solitons & Fractals* 128 (2019) 355–365.
 - [33] S. Kumar, A. Kumar, B. Samet, J.F. Gómez-Aguilar, M.S. Osman, A chaos study of tumor and effector cells in fractional tumor-immune model for cancer treatment, *Chaos, Solitons & Fractals* 141 (2020) 110321.
 - [34] S. Kumar, R. Kumar, J. Singh, K.S. Nisar, D. Kumar, An efficient numerical scheme for fractional model of HIV-1 infection of CD4+ T-cells with the effect of antiviral drug therapy, *Alex. Eng. J.* 59 (4) (2020) 2053–2064.
 - [35] A. Atangana, A. Akgül, Can transfer function and Bode diagram be obtained from Sumudu transform, *Alex. Eng. J.* 59 (4) (2020) 1971–1984.
 - [36] A. Atangana, Ali Akgül, K.M. Owolabi, Analysis of fractal fractional differential equations, *Alex. Eng. J.* 59 (3) (2020) 1117–1134.
 - [37] A. Atangana, A. Akgül, On solutions of fractal fractional differential equations, *Dis. Cont. Dyn. Syst.S* (2018), <https://doi.org/10.3934/dcdss.2020421>.
 - [38] M. Rafei, H. Daniai, D.D. Ganji, Variational iteration method for solving the epidemic model and the prey and predator problem, *Appl. Math. Comput.* 186 (2007) 1701–1709.
 - [39] H. Haq, K. Shah, G. Rahman, M. Shahzad, Numerical solution of fractional order smoking model via laplace adomian decomposition method, *Alex. Eng. J.* 57 (2) (2018) 1061–1069.
 - [40] F. Haq, K. Shah, G. Rahman, M. Shahzad, Numerical analysis of fractional order model of HIV-1 infection of CD4+ t-cells, *Comput. Meth. Differ. Equ.* 5 (1) (2017) 1–11.
 - [41] S. Kumar, R. Kumar, M.S. Osman, B. Samet, B.A wavelet based numerical scheme for fractional order SEIR epidemic of measles by using Genocchi polynomials, *Numer. Methods Partial Diff. Eqs.* (2020), <https://doi.org/10.1002/num.22577>.
 - [42] S. Kumar, K.S. Nisar, R. Kumar, C. Cattani, B. Samet, A new Rabotnov fractional-exponential function-based fractional derivative for diffusion equation under external force, *Math. Methods Appl. Sci.* 43 (7) (2020) 4460–4471.
 - [43] S. Kumar, S. Ghosh, R. Kumar, M. Jleli, A fractional model for population dynamics of two interacting species by using spectral and Hermite wavelets methods, *Numer. Methods Partial Differential Eqs.* (2020), <https://doi.org/10.1002/num.22602>.
 - [44] A.S. Shaikh, I.N. Shaikh, K.S. Nisar, A mathematical model of covid-19 using fractional derivative: Outbreak in india with dynamics of transmission and control, *Adv. Diff. Equ.* 2020 (2020) 373.

- [45] A. Khan, T.S. Khan, M.I. Syam, H. Khan, Analytical solutions of time-fractional wave equation by double laplace transform method, *Eur. Phys. J. Plus* 134 (4) (2019) 163.
- [46] S. Kumar, A. Kumar, B. Samet, H. Dutta, A study on fractional host-parasitoid population dynamical model to describe insect species, *Numer. Methods Partial Differential Eqs.* (2020), <https://doi.org/10.1002/num.22603>.
- [47] M. Kaplan, B. Ahmet, A novel analytical method for time-fractional differential equations, *Optik* 127 (20) (2016) 8209–8214.
- [48] M. Caputo, M. Fabrizio, A new definition of fractional derivative without singular kernel, *Prog. Fract. Diff. Appl.* 1 (2) (2015) 73–85.
- [49] D. Baleanu, A. Jajarmi, H. Mohammadi, S. Rezapour, A new study on the mathematical modelling of human liver with caputo-fabrizio fractional derivative, *Chaos Soliton & Fractals* 134 (2020) 109705.
- [50] T.A. Yildiz, A. Jajarmi, B. Yildiz, D. Baleanu, New aspects of time fractional optimal control problems within operators with nonsingular kernel, *Discr. Contin. Dyn. Syst.* 13 (3) (2020) 407.
- [51] K.M. Owolabi, A. Atangana, A. Akgül, Modelling and analysis of fractal-fractional partial differential equations: Application to reaction-diffusion model, *Alex. Eng. J.* 59 (4) (2020) 2477–2490.
- [52] A. Atangana, D. Baleanu, New fractional derivatives with non-local and non-singular kernel: theory and application to heat transfer model, *Therm. Sci.* 20 (2) (2016) 763–785.
- [53] A. Atangana, Non validity of index law in fractional calculus: a fractional differential operator with markovian and non-markovian properties, *Physica A: Stat. Mech. Appl.* 505 (2018) 688–706.
- [54] M. ur Rahman, M. Arfan, K. Shah, J.F. Gómez-Aguilar. Investigating a nonlinear dynamical model of COVID-19 disease under fuzzy caputo, random and ABC fractional order derivative, *Chaos, Solitons & Fractals*, 140 (2020) 110232.
- [55] S. Ahmad, A. Ullah, M. Arfan, K. Shah, On analysis of the fractional mathematical model of rotavirus epidemic with the effects of breastfeeding and vaccination under Atangana-Baleanu (AB) derivative, *Chaos, Solitons & Fractals* 140 (2020) 110233.
- [56] S. Kumar, R.P. Chauhan, S. Momani, S. Hadid, Numerical investigations on COVID-19 model through singular and non-singular fractional operators, *Numer. Methods Partial Differential Eqs.* (2020), <https://doi.org/10.1002/num.22707>.
- [57] S. Kumar, R. Kumar, R.P. Agarwal, B. Samet, A study of fractional Lotka-Volterra population model using Haar wavelet and Adams-Bashforth-Moulton methods, *Math. Methods Appl. Sci.* 43 (8) (2020) 5564–5578.
- [58] K. Logeswari, C. Ravichandran, K.S. Nisar, Mathematical model for spreading of COVID-19 virus with the Mittag-Leffler kernel, *Numer. Methods Partial Differential Eq.* 2020 (2020) 1–16.
- [59] A. Akgül, A novel method for a fractional derivative with non-local and non-singular kernel, *Chaos, Solitons & Fractals* 114 (2018) 478–482.
- [60] A. Naghipour, J. Manafian, Application of the Laplace adomian decomposition method and implicit methods for solving Burger's equation, *J. Pure. Appl. Math.* 6 (1) (2015) 68–77.
- [61] S.Z. Rida, A.S. Abdel Rady, A.A.M. Arafa, M. Khalil, Approximate analytical solution of the fractional epidemic model, *IJMR* 1 (2012), 17–19.
- [62] S.C. Brailsford, P.R. Harper, B. Patel, M. Pitt, An analysis of the academic literature on simulation and modelling in health care, *J. Simul.* 3 (3) (2009) 130–140.
- [63] J. Rappaz, R. Touzani, On a two-dimensional magnetohydrodynamic problem: modelling and analysis, *Math. Model. Numer. Anal.* 26 (2) (1992) 347–364.
- [64] G. Wong, W. Liu, Y. Liu, B. Zhou, Y. Bi, G.F. Gao, MERS, SARS, and Ebola: the role of super-spreaders in infectious disease, *Cell Host Microbe* 18 (4) (2015) 398–401.
- [65] P. Van den Driessche, J. Watmough, Reproduction numbers and sub-threshold endemic equilibria for compartmental models of disease transmission, *Math. Biosci.* 180 (1–2) (2002) 29–48.
- [66] S. Kumar, R. Kumar, C. Cattani, B. Samet, Chaotic behaviour of fractional predator-prey dynamical system, *Chaos, Solitons & Fractals* 135 (2020) 109811.